Discussion

Tuesday, October 20, 2020 6:50 PM

Ch 1.3 − 1.5 (2.1)

(Ex) How many 14 letter words can we create using the letters of "UCLA ASHE CENTER" (w/out repetition)?

A) $14!$ B) $\frac{14!}{2}$ C) $\frac{14!}{12}$ D) $\frac{14!}{24}$ E) $\frac{14!}{120}$

Sol: $U C_1 L A_1 A_2 S H E_1 C_2 E_2 N T E_3 R \to 14!$

$E_1 E_2 U C_2 H C_1 E_3 N \ldots$

$E_3 E_1 U C_2 H C_1 E_2 N \ldots$

order A_1 & A_2 & A_2 changed.

$\downarrow$

$E E U C H C E . N \ldots$

$3! = 6$ different permutatis of E's will give the same results

$2! = 2$ " " " A_s " " " "

$2! = 2$ " " " C_s " " " "

$6 \times 2 \times 2 = 24$ different permutations (among 14! permutations) actually gives the same output.

$E_1 E_2 E_3 A B C \to 6!$

$\downarrow$

$A E_1 B C E_2 E_3$

$A E_1 B C E_3 E_2$

$A E_2 B C E_1 E_3$

$A E_2 B C E_3 E_1$

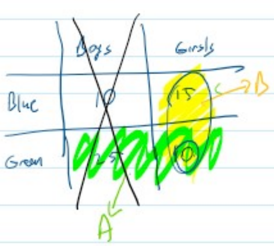
$A E_3 B C E_1 E_2$

$A E_3 B C E_2 E_1$

$\to$ AEBCEE.

$\frac{6!}{3!} = \frac{6!}{6}$

Ch 1.3 (Conditional probability)



The teacher chooses student S. S he wants the students to guess the eye color of S.

$A = \{S$ has green eyes$\}$

$P(A) = \frac{35}{60} > \frac{1}{2}$

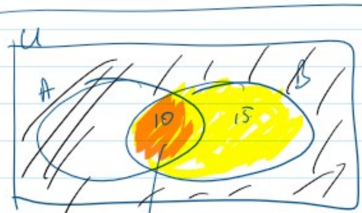
Then teachers says S is girl.

$B = \{S$ is a girl$\}$.

$P(A|B) = P($S has green eyes given S is a girl$) = \frac{10}{25}$

$\hookrightarrow P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{10}{60}}{\frac{25}{60}} = \frac{10}{25}.$

Another way of thinking.



$P(A|B) = \frac{P(A \cap B)}{P(B)}$

$\hookrightarrow$ within B, $A \cap B$ is where A happens

$P(A|B) = P(B|A)$?

$$P(A|B) = \frac{P(A\cap B)}{P(B)} = \frac{10}{25}$$ ← Not sure.

A → S has green eyes; B → S is girl!

	Boys	Girls
Blue	10	15
Green	25	10

$$P(B|A) = \frac{P(A\cap B)}{P(A)} = \frac{\frac{10}{60}}{\frac{35}{60}} = \frac{10}{35}$$

A: Student has green eye.

Ex: Monica throws two dice in a backgammon game. You know that the sum of dice is 10. What is the probability that one of the dice is 6?

A) $\frac{1}{2}$ B) $\frac{1}{3}$ C) $\frac{2}{3}$ D) 1 E) 0

Sol:

D_1	D_2
→ 4	6
→ 6	4
5	5

→ Because of given information → $\frac{2}{3}$

A = { One of the dice is 6 }

B = { Sum of 2 dice is 10 }

$$P(A|B) = \frac{P(A\cap B)}{P(B)} = \frac{\frac{2}{36}}{\frac{3}{36}} = \frac{2}{3}$$

(Ex 1.5.9 - modified) There is a new diagnostic test for covid19 that occurs about 10% of the population. The test is not perfect, but will detect a person with disease 95% of time. It will, however, say that a person without the disease has the disease about 20% of time. A person selected at random from the population, the test says this person has the disease. What is the conditional prob. that this person actually has the disease?

A) $\frac{1}{3}$ B) $\frac{9}{100}$ $\boxed{C)\ \frac{19}{55}}$ D) $\frac{15}{66}$ E) $\frac{33}{95}$

Sol: Let S be the randomly selected person.

D = { S has disease }

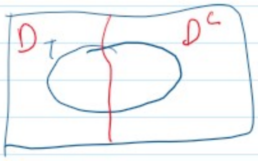
T = { The test says S has the disease }

⇒ $P(D|T)$

$P(D) = 0.1$ $P(T|D) = 0.95$ $P(T|D^c) = 0.2$

$$P(D|T) = \frac{P(D\cap T)}{P(T)} = \frac{P(D\cap T)}{\underbrace{P(T\cap D)+P(T\cap D^c)}_{P(T)}} = \frac{\overbrace{P(D\cap T)}^{0.95\times 0.1}}{P(T|D)\cdot P(D) + P(T|D^c)P(D^c)} = \frac{0.95\times 0.1}{0.95\times 0.1 + 0.2\times 0.9} = \boxed{\frac{19}{55}}$$

$P(T|D) = 0.95$, $P(D) = 0.1$, $P(T|D^c) = 0.2$, $P(D^c) = 1-P(D) = 0.9$



$P(T|D) = 0.95$

$$P(T|D) = \frac{P(T\cap D)}{P(D)} \Rightarrow P(T\cap D) = P(T|D)\cdot P(D)$$

$P(T|D) = 0.95.$

$$P(T|D) = \frac{P(T\cap D)}{P(D)} \Rightarrow P(T\cap D) = P(T|D)\cdot P(D)$$